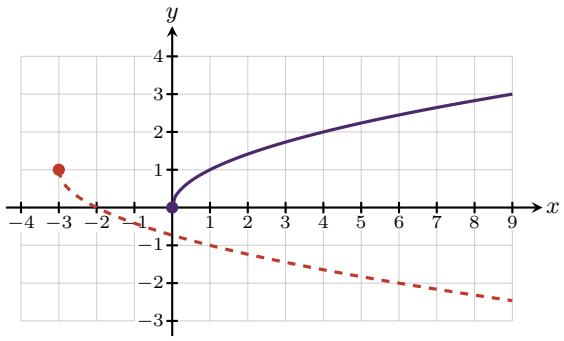
Instructions: Show all work in the space provided. Write each final answer in its box. Only what is inside a box is graded. Write the letter of the single best answer in its Letter box. Write intervals in interval notation. No notes. **70 points total.**

Part A — Transformations (12 pts)

1. The solid graph is $f(x) = \sqrt{x}$. The dashed graph is a transformation of f. Fill in the boxes so the equation matches the dashed graph.

$$g(x) = \boxed{-1}\, f\left(\boxed{1}\, x + \boxed{3}\right) + \boxed{1}$$



2. The graph of $y = 3\sqrt{x-2}$ is the graph of the parent $y = \sqrt{x}$ after transformations. Complete the table: where does each landmark point of the parent land?

Parent point on $y = \sqrt{x}$	$(0, 0)$	$(1, 1)$	$(4, 2)$
Lands at on $y = 3\sqrt{x-2}$	$(2, 0)$	$(3, 3)$	$(6, 6)$

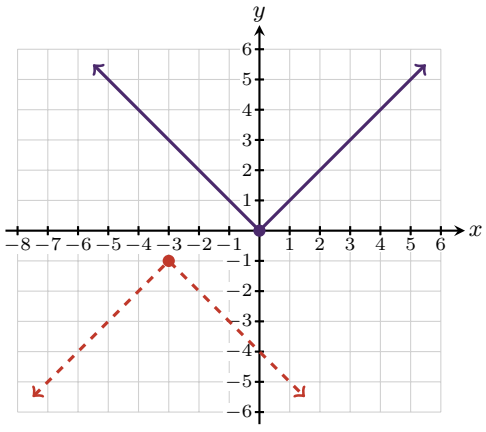
3. The graph of $y = \left|\frac{1}{4}x\right| - 3$ is the graph of the parent $y = |x|$ after transformations. Complete the table.

Parent point on $y = \lvert x \rvert$	$(-1, 1)$	$(0, 0)$	$(1, 1)$
Lands at on $y = \left\lvert\frac{1}{4}x\right\rvert - 3$	$(-4, -2)$	$(0, -3)$	$(4, -2)$

4. The solid graph is $y = |x|$. Which function is the dashed graph?

(A) $y = -|x - 3| - 1$
(B) $y = |x + 3| - 1$
(C) $y = -|x + 3| - 1$
(D) $y = -|x + 1| - 3$

Letter: **C**



Part B — Symmetry, One-to-One, and Inverses (14 pts)

5. Fill in each table of values. Then write whether the function is *even*, *odd*, or *neither*.

$f(x) = x^3 - 3x$

x	y
-2	-2
-1	2
0	0
1	-2
2	2

Odd

$g(x) = 2|x| - 1$

x	y
-2	3
-1	1
0	-1
1	1
2	3

Even

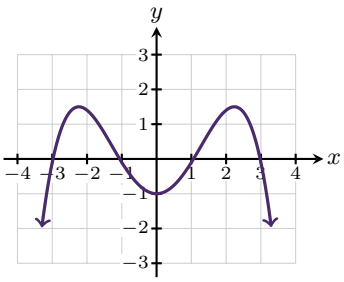
$h(x) = x^2 + x$

x	y
-2	2
-1	0
0	0
1	2
2	6

Neither

For 6–8, choose from the answers under each graph.

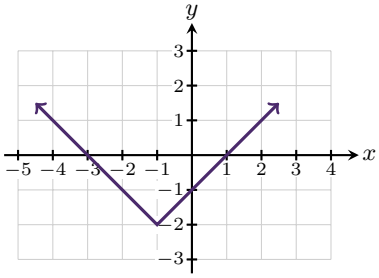
6. Is the function even, odd, or neither?



(A) even (B) odd (C) neither

Letter: **A**

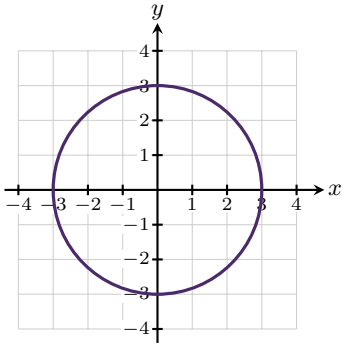
7. Use the vertical and horizontal line tests. The graph shows



(A) a one-to-one function
(B) a function, not one-to-one
(C) not a function

Letter: **B**

8. Use the vertical and horizontal line tests. The graph shows



(A) a one-to-one function
(B) a function, not one-to-one
(C) not a function

Letter: **C**

9. Let $f(x) = \sqrt[3]{3x+1}$. Which function is $f^{-1}(x)$?

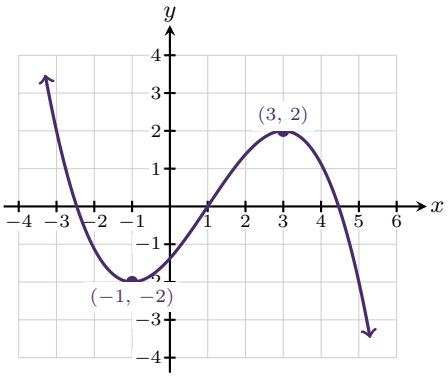
$$y = \sqrt[3]{3x+1}$$
$$x = \sqrt[3]{3y+1}$$
$$x^3 = 3y + 1$$
$$y = \frac{x^3 - 1}{3}$$

(A) $\dfrac{x^3+1}{3}$ (B) $\dfrac{x^3}{3} - 1$ (C) $\dfrac{x^3-1}{3}$ (D) $\dfrac{1}{\sqrt[3]{3x+1}}$

Letter: **C**

Part C — Reading Characteristics (21 pts)

10. The graph of f continues in the direction of its arrows. Its turning points are marked.



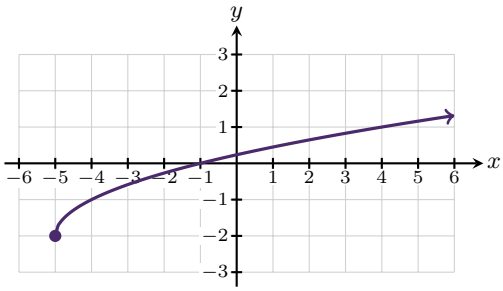
Increasing on:

$(-1, 3)$

Decreasing on:

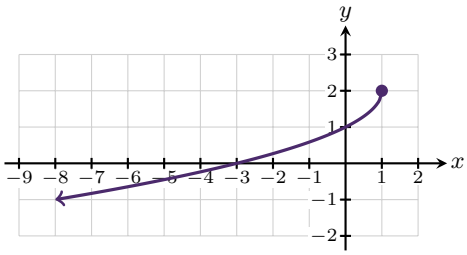
$(-\infty, -1) \cup (3, \infty)$

11. State the domain of the function graphed.



Domain: $[-5, \infty)$

12. State the range of the function graphed.



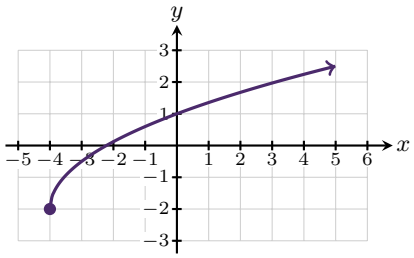
Range: $(-\infty, 2]$

13. Find the domain of $f(x) = \sqrt{3x+12} - 4$ algebraically.

$$3x + 12 \geq 0$$
$$3x \geq -12$$
$$x \geq -4$$

Domain: $[-4, \infty)$

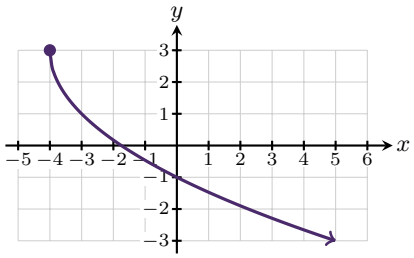
14. Is the graph concave up or concave down?



(A) concave up (B) concave down

Letter: **B**

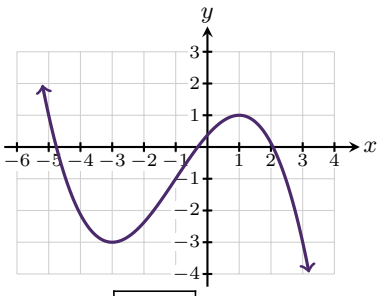
15. Is the graph concave up or concave down?



(A) concave up (B) concave down

Letter: **A**

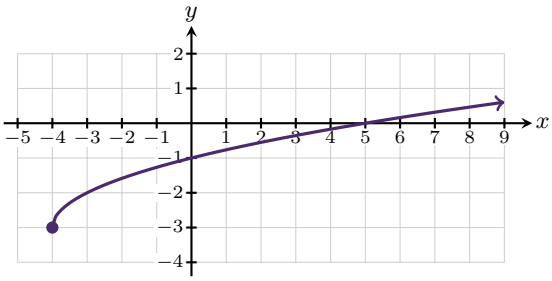
16. Find the point(s) of inflection.



How many? 1

At: $(-1, -1)$

17. The function f is graphed at the right.



(a) Domain of f: $[-4, \infty)$

(b) Range of f: $[-3, \infty)$

(c) Which sentence is true?

(A) f is not invertible because it is not one-to-one.

(B) f is invertible because it passes the vertical line test.

(C) f is invertible because it passes the horizontal line test.

Letter: **C**

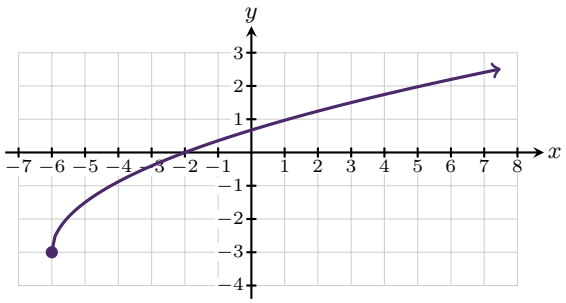
(d) Domain of f^{-1}: $[-3, \infty)$

(e) Range of f^{-1}: $[-4, \infty)$

18. Which type of function is graphed?

(A) exponential
(B) absolute value
(C) cube root
(D) square root
(E) linear

Letter: **D**



Part D — Average Rate of Change and Secant Lines (6 pts)

Let $h(x) = \sqrt[3]{x+1} - 2$.

$$\begin{aligned} h(-9) &= \sqrt[3]{-8} - 2 \\ &= -4 \\ h(7) &= \sqrt[3]{8} - 2 \\ &= 0 \\ \frac{h(7) - h(-9)}{7 - (-9)} &= \frac{0 - (-4)}{16} \\ &= \frac{1}{4} \\ y - 0 &= \tfrac{1}{4}(x - 7) \\ y &= \tfrac{1}{4}x - \tfrac{7}{4} \end{aligned}$$

19. Find the average rate of change of h from $x = -9$ to $x = 7$. Write your answer as a simplified fraction.

1/4

20. Find the secant line of h from $x = -9$ to $x = 7$, in slope-intercept form. $y =$ $\frac{1}{4}x - \frac{7}{4}$

Part E — One Function, Every Characteristic (17 pts)

Problems 21–27 are all about the function

$$g(x) = 2\,|x + 1| - 6.$$

21. Describe the transformations that take $y = |x|$ to g.

Shift left 1, stretch vertically by a factor of 2, then shift down 6.

22. The domain of g is $(-\infty, \infty)$, and the range is $[-6, \infty)$.

23. g is increasing on $(-1, \infty)$ and decreasing on $(-\infty, -1)$.

24. Find every intercept of g.

$$\begin{aligned} 2|x+1|-6 &= 0 \\ |x+1| &= 3 \\ x+1 &= 3 \text{ or } x+1 = -3 \\ x &= 2 \text{ or } x = -4 \\ g(0) &= 2|0+1|-6 \\ &= -4 \end{aligned}$$

x-intercept(s): $x =$ $-4,\ 2$ y-intercept: $(0, -4)$

For 25 and 26, choose from these four statements.

(A) $\lim_{x\to\infty} g(x) = \infty$ (B) $\lim_{x\to-\infty} g(x) = \infty$ (C) $\lim_{x\to\infty} g(x) = -\infty$ (D) $\lim_{x\to-\infty} g(x) = -\infty$

25. The left end behavior of g: *Letter:* **B**

26. The right end behavior of g: *Letter:* **A**

27. Complete the sentence: the **minimum** (*maximum* or *minimum*) value of g is -6, and it occurs at $x =$ -1.